Team Member Contributions (Group 7)

The Project we are doing is Password Cracking and Strength Evaluation Analysis of the GroceryGo Web Application. We are 3 team members and there are 5 phases for our group Project. We have approximately 9 weeks for the final presentation for this project. So I will create a timeline and contribution list of each member of this project group:

1. Sangram Mathews - Attack & Environment Lead

Responsibilities:

- Configure Linux VM
- Setting up shared GitHub repository
- Install Hashcat / John the Ripper
- Create SQLite password database
- Export sample hashed password file
- Extract hashes
- Perform dictionary attack
- Record cracking time
- Measure time complexity
- Write Attack Implementation section

Contribution:

- Designing and executing the offensive security components of the project
- Setting up Lab environment
- Creating SQL database with weak passwords for users
- Hashing and performing dictionary attack on 1st round
- Performing Last round of Cracking

2. Ayman Fahim – Attack and Secure Implementation Lead

- **Responsibilities:**
  - Leading the Encryption Processes
  - Implement SHA-256 password storage (Python simulation)
  - Integrate hashing into GroceryGo test environment
  - Validate hashing correctness
  - Document security weakness
  - Implement per-user random salt
  - Rehash passwords
  - Attempt the 2nd round of dictionary attack
  - Replace SHA-256 with bcrypt
  - Store bcrypt hashes

- **Contribution:**
    - Strengthening the authentication system of GroceryGo by implementing and comparing different password storage mechanisms.
    - Hashing and performing the 2^{nd} Round of Cracking
    - Developed the SHA-256 hashing baseline, integrated per-user salting, and later replaced the hashing scheme with bcrypt to demonstrate adaptive hashing security.

3. Maciej Leniartek- Testing, Evaluation & Documentation Lead

- **Responsibilities:**
    - Design experiment tracking sheet (Excel/CSV)
    - Creating the list of weak passwords for 40 Users
    - Keeping records of password cracking time
    - Analyze success rate
    - Create baseline comparison table
    - Create visual charts
    - Analyze computational cost
    - Create comparison matrix: SHA-256, SHA-256 + Salt, bcrypt
    - Running assurance tests
- **Contribution:**
    - Leading Password Strength Evaluation
    - Documentation and Presentation
    - Measure resistance levels
    - Write Results, Graphs, Discussion, Conclusion
    - Format references (NIST, OWASP, bcrypt paper)

Project Timeline & Task Assignment

Week	Phase	Tasks	Lead Member	Deliverable
Week 1	Planning & Research	Finalize scope, define metrics, review OWASP & NIST guidelines, assign responsibilities	All Members	Project roadmap & lab design outline
Week 2	Phase 1 – Lab Setup	Configure Linux VM, install Hashcat/John, create SQLite test database	Member 1	Functional cracking lab environment

Week	Phase	Tasks	Lead Member	Deliverable
Week 3	Phase 1 – Hashing Setup	Implement SHA-256 password storage, prepare test dataset	Member 2	Baseline hashing system
Week 4	Phase 2 – Baseline Attack	Extract unsalted SHA-256 hashes, perform dictionary attack, record cracking time	Member 1	Baseline cracking results report
Week 5	Phase 3 – Add Salting	Implement per-user salt, rehash passwords, repeat cracking test	Member 2	Salted vs unsalted comparison data
Week 6	Phase 4 – Implement bcrypt	Replace SHA-256 with bcrypt, configure cost factor, test cracking resistance	Member 2	bcrypt security evaluation report
Week 7	Phase 5 – Password Strength Testing	Enforce complexity rules, test weak vs strong passwords, analyze success rates	Member 3	Password strength impact analysis
Week 8	Documentation & Analysis	Create charts, comparison tables, write Results, Discussion & Conclusion	Member 3	Complete draft of final report
Week 9	Presentation Preparation	Build cybersecurity-themed slides, add graphs & screenshots, rehearse presentation	All Members	Final presentation